

RAG (Retrieval Augmented Generation) - Basic Flow

1. Load docs → PyPDFLoader (Load PDF documents)
2. Split → RecursiveCharacterTextSplitter (Chunking - break text into smaller parts)
3. Embed → OpenAIEmbeddings (Convert text into vectors)
4. Store → FAISS (Store embeddings in vector database)
5. Retrieve → Retriever / as_retriever() (Fetch relevant chunks)
6. Context Injection → RunnablePassthrough (Add retrieved data as context)
7. Prompt Structuring → ChatPromptTemplate (Combine context + question)
8. Send to LLM → ChatOpenAI (Generate response)
9. Output Parsing → StrOutputParser (Format final output)

Flow Summary:

Load → Split → Embed → Store → Retrieve → Context → Prompt → LLM → Output